

# Hyung-gun Chi

RESEARCH SCIENTIST

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## Summary

Research Scientist building **real-time multimodal foundation models for wearable AI and smart glasses**. Track record from 3D human activity understanding (**InfoGCN**) to speech/audio foundation model compression (**DiceHuBERT**) to on-device LLMs for **Ray-Ban Meta glasses**. PhD from Purdue; research & engineering at Google Android XR, Meta Reality Labs, and Apple AIML.

## Professional Experience

### Google – Multimodal AI for Smart Glasses

San Jose, CA, USA

RESEARCH SCIENTIST

2026 - Present

- Designing and deploying **multimodal foundation models** across audio, vision, sensor, and text for AI-powered smart glasses at **Android XR**.
- Advancing real-time wearable AI through on-device model architectures, multimodal LM adaptation, and prototype-to-production applied research.

### Meta Reality Labs – Voice AI

Menlo Park, CA, USA

RESEARCH ENGINEER

2025 - 2026

- Led **on-device LLMs** for **Ray-Ban Meta glasses** and **Ray-Ban Meta Display** under strict power, memory, and latency budgets.
- Shipped on-device smart reply on mobile, improving data coverage through synthetic data generation and cross-team pipeline integration.
- Delivered hardware-aware optimizations, including 4-bit W/A quantization, KV-cache management, and attention/operator tuning, across a monthly end-to-end deployment cadence.

### Apple – Siri Speech & Understanding

Cupertino, CA, USA

AIML RESIDENT / MACHINE LEARNING ENGINEER

2024 - 2025

- Developed **multimodal perception models** (gesture, video, audio) and speech/audio foundation model compression methods for contextual agents.
- Proposed **DiceHuBERT**, an architecture-agnostic self-supervised distillation for speech foundation models; +21% ASR and +14% PR on SUPERB over prior KD (Interspeech 2025).
- Designed adaptive knowledge distillation for on-device device-directed speech detection, improving robustness under noisy conditions (Interspeech 2025).

### Toyota Research Institute

Los Altos, CA, USA

ML RESEARCH INTERN

Summer 2023

- Designed a novel **vision-language-tactile representation** for robotic perception, improving manipulation accuracy.
- Curated and released a large-scale multimodal robotic dataset (IROS 2024).

### Honda Research Institute

San Jose, CA, USA

CV RESEARCH INTERN

Spring 2023 & Summer 2022

- Developed **discrete diffusion models** for text-to-human-motion generation, achieving SOTA smoothness and realism (ECCV 2024).
- Researched Neural-ODE-based Transformers for trajectory prediction and future action forecasting (CVPR 2024).

### Convergence Design Lab, Purdue University

West Lafayette, IN, USA

GRADUATE RESEARCH ASSISTANT

2018 - 2023

- Pioneered **3D geometric deep learning** and vision-language representation for human activity understanding.
- Developed InfoGCN / InfoGCN++, SOTA skeleton action recognition models (CVPR 2022, TPAMI 2024), cited 500+ times.
- Published 15+ papers (CVPR, ECCV, TPAMI, ICRA) and filed 7+ patents in multimodal AI.

### Heum Labs

Seoul, South Korea

CO-FOUNDER & CEO

2016 - 2017

- Co-founded and led a 30-person startup developing tax and accounting automation software for small business owners.

## Education

### Purdue University

West Lafayette, IN, USA

PHD IN ELECTRICAL AND COMPUTER ENGINEERING

2023

- **Thesis:** Towards Improved Representations on Human Activity Understanding
- **Advisor:** Dr. Karthik Ramani

### Purdue University

West Lafayette, IN, USA

MS IN ELECTRICAL AND COMPUTER ENGINEERING

2022

### Yonsei University

Seoul, South Korea

BS IN MECHANICAL ENGINEERING

2017

## Selected Publications / Patents

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### Publications (1,000+ citations, h-index 13)

- [15] *DiceHuBERT: Distilling HuBERT with a Self-Supervised Learning Objective*  
H Chi, Z Aldeneh, T Likhomanenko, O Rudovic, T Higuchi, L Chen, S Watanabe, A Abdelaziz, **Interspeech**, 2025.
- [14] *Adaptive Knowledge Distillation for Device-Directed Speech Detection*  
H Chi, F Pesce, W Chang, O Rudovic, A Argueta, S Barun, V Garg, A Abdelaziz, **Interspeech**, 2025.
- [13] *CARING-AI: Towards Authoring Context-aware Augmented Reality INstruction through Generative Artificial Intelligence*  
J Shi, R Jain, S Chi, H Doh, **H Chi**, AJ Quinn, K Ramani, **CHI**, 2025.
- [12] *Context-Enriched Voxel Queries for Camera-based 3D Occupancy Prediction*  
G Oh, S Kim, H Ko, **H Chi**, J Kim, D Lee, D Ji, S Choi, S Jang, S Kim, Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
- [11] *InfoGCN++: Learning Representation by Predicting the Future for Online Skeleton-based Action Recognition*  
**H Chi**\*, S Chi\*, Q Huang, K Ramani, Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2024.
- [10] *M2D2M: Discrete Diffusion Model for the Multi-Motion Generation from the Text*  
**H Chi**\*, S Chi\*, H Ma, N Agarwal, F Siddiqui, K Ramani, K Lee, European Conference on Computer Vision (**ECCV**), 2024.
- [9] *Enhanced Motion Forecasting with Visual Relation Reasoning*  
S Kim, H Baek, S Lee, **H Chi**, H Lim, J Kim, S Kim, European Conference on Computer Vision (**ECCV**), 2024.
- [8] *VisionTrap: Vision-Augmented Trajectory Prediction Guided by Textual Descriptions*  
S Moon, H Woo, H Park, H Jung, R Mahjourian, **H Chi**, H Lim, S Kim, J Kim, European Conference on Computer Vision (**ECCV**), 2024.
- [7] *Multi-Modal Representation Learning with Tactile Modality*  
**H Chi**\*, J Barreiros\*, J Mercat, K Ramani, T Kollar, International Conference on Intelligent Robots and Systems (**IROS**), 2024.
- [6] *Higher-order Relation Reasoning for Trajectory Prediction*  
S Kim, **H Chi**, H Lim, K Ramani, J Kim, S Kim, Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
- [5] *AdamsFormer for Spatial Action Localization in the Future*  
**H Chi**, K Lee, N Agarwal, Y Xu, K Ramani, C Choi, Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.
- [4] *Uncovering the Missing Pattern: Unified Framework Towards Trajectory Imputation and Prediction*  
X Yi, A Bazarjani, **H Chi**, C Choi, Y Fu, Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.
- [3] *Pose Relation Transformer: Refine Occlusions for Human Pose Estimation*  
**H Chi**\*, S Chi\*, C Stanley, K Ramani, International Conference on Robotics and Automation (**ICRA**), 2023.
- [2] *InfoGCN: Representation Learning for Human Skeleton-based Action Recognition*  
**H Chi**\*, MH Ha\*, S Chi, SW Lee, Q Huang, K Ramani, Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022.
- [1] *A Large-scale Annotated Mechanical Components Benchmark for Classification and Retrieval Tasks with Deep Neural Networks*  
**H Chi**\*, S Kim\*, X Hu, Q Huang, K Ramani, European Conference on Computer Vision (**ECCV**), 2020.

### Patents (2 granted, 5 pending)

- [P7] *Multi-Motion Generation*,  
S Chi, **H Chi**, F Siddiqui, N Agarwal, H Ma, K Lee, US Patent App.
- [P6] *System and Method for Authoring Context-augmented Reality Instruction through Generative Artificial Intelligence*,  
K Ramani, S Chi, **H Chi**, R Jain, J Shi, US Patent App.
- [P5] *Pose Relation Transformer Refine Occlusions for Human Pose Estimation*,  
K Ramani, **H Chi**, S Chi, US Patent App. 18/584,191.
- [P4] *Spatio Action Localization in the Future*,  
**H Chi**, K Lee, N Agarwal, Y Xu, C Choi, US Patent App. 18/300,844.
- [P3] *Trajectory Imputation and Prediction*,  
Y Xu, A Bazarjani, **H Chi**, C Choi, US Patent App. 18/182,195.
- [P2] *Pixel-wise Hand Segmentation of Multi-modal Hand Activity Video Dataset*,  
K Ramani, S Kim, **H Chi**, US Patent 11,562,489.
- [P1] *Computer Input System for Office/Factory Automation*,  
**H Chi**, WO Patent 2018/074729 A1.

## Academic Services & Recognition

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### Reviewer

- Conferences: **CVPR**, **ICCV**, **ECCV**, **NeurIPS**, **ICML**, **ICLR**, **AAAI**, **CHI**, **UIST**, **BMVC**, **ACCV**, **ICRA**, **IROS**, **ICPR**, **ICASSP** (2021-present).
- Journals: **TPAMI**, **IJCV**, **TIP**, **PR**, **TNNLS**, **CVIU**, **JVCI**, **TOMM**, **R-AL**, **TETCI**, **JCISE**.

### Invited Talks

- *Advancements in Human Action Recognition by Learning Skeleton Representations* – Sungkyunkwan University, Seoul (May 2023).
- *Representation Learning for Recognizing Human Activity* – Yonsei University & Keimyung University (Apr 2023).
- *Learning Representation for Human Action Recognition* – Hongik University, Seoul (Dec 2022).

### Recognition

- Invited participant, **CVPR 2024 Doctoral Consortium** (Seattle, WA).